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RESEARCH ARTICLE

75MW AC PV Module Field Anomaly Detection Using Drone-Based IR Orthogonal Images With Res-CNN3 Detector

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ABSTRACT In recent years, the use of solar photovoltaic (PV) technologies for generating electricity has gained much popularity due to their efficiency, cost-effectiveness, reliability and the need to reduce carbon emissions and air pollution levels around the world in order to control and limit global climate change. Drone-based inspection of Solar Plants is an efficient method to perform preventative and corrective maintenance on the Solar PV arrays installed in large-scale grid-connected Solar PV Plants. Quite a few studies have been made in detecting PV module anomalies from these drone-captured thermal images. The main challenge that most of the applied techniques and models in these studies experience is the capability to detect and localize various PV module defects with high and consistent confidence scores and accuracy. The efficiency and robustness of anomaly detection of these models are also reduced when the testing image scales vary. In this article, we propose the Res-CNN3 framework for defective PV modules region object detection. Res-CNN3 follows a bipartite process for defect regions detection to process the thermal delta of a thermal PV module image using a combination of concatenated CNNs and residual networks to identify whether the thermal image is characteristic of defect regions or not. Logistic regression is employed as the loss function whilst the Selective Search (SS) algorithm is utilized to predict the regions of interest (RoI) of the input thermal images for PV module defect regions object detection. Several categories of experiments are performed on the consolidated dataset. The results of these experiments demonstrate that the proposed model surpasses the advanced benchmarked methods in accuracy and computing resource efficiency.

INDEX TERMS Deep learning, defects, PV module, solar, drone.

I. INTRODUCTION

The constant increase in the number of installed large-scale solar photovoltaic PV plants has opened the Solar PV industry to new challenges and opportunities for effective operations and maintenance strategies for the solar PV arrays. At the same time, new findings about solar PV module performance, behavior and ageing under various climatic and operational conditions are on the rise. Among these findings, is the nonlinear performance degradation of PV modules due to various defects that sporadically develop over the lifetime of the product [1]. Some of these PV module defects are visible to the naked eye as they tend to occur on the mechanical

structure of the PV module. Other PV module defects tend to take on a much rather discrete characteristic as they occur on the elemental level of the modules affecting the PV module cells, encapsulants, etc and are much more arduous to discern with the naked eye. In both scenarios of defects, abnormal operation of the individual PV modules is induced, which affects the PV arrays, and the electrical output of the complete PV plant is reduced [1], [2].

When conducting PV maintenance inspections to replace underperforming or defective PV modules for improving Plant electrical output, it can be a very time-consuming, unsafe, labor-intensive and at times cumbersome exercise to perform manual inspections of the many solar PV arrays installed in large-scale Solar PV plants, which often comprise hundreds of thousands of PV modules in total. These

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challenges are increasingly being addressed by performing drone-based inspections of the Solar PV arrays [3].

Drone-based inspections of the Solar PV arrays offer two main advantages: Better health, safety and environmental measures for the inspectors and efficiency in operations and maintenance regimes. The drone or unmanned aerial vehicle (UAV) is generally equipped with a thermal camera to perform a contactless inspection of the PV plant by capturing aerial images of the solar PV arrays during flight time. However, as some thermal cameras have a low resolution and the UAVs fly at high airspeed, the main limitation of this type of inspection approach is that it can only detect the hot-spots with large sizes and high temperatures given the applied camera sensor technologies. Conversely, hot-spots are normally a result of various defects, such as residual stresses, minor cracks, burn marks, internal short-circuits, breakages, delamination and several others. Generally, these types of defects are more arduous to differentiate by thermographic inspections, since infrared, thermographic or pseudo-colour images only contain thermal radiation or thermal delta information [4].

The way to resolve these constraints or shortcomings of drone-based thermographic inspections is to apply visual assessment by using a drone assembled with a high-definition (HD) photographic camera sensor. The camera sensor will then detect small-size defects by capturing RGB and orthogonal images. These true-color images from the HD camera-based visual inspections are more discriminative and precise than thermographic inspections for the recognition of defects such as delamination, short-circuits, corrosion, module breakages, snail tails, discoloration, and dirty points [5].

By further applying deep learning object recognition or detection techniques as part of the data processing on these RGB and orthogonal images, an efficient PV module anomaly detector can be designed. This approach of PV module anomaly detection has been presented in various research works recently and has demonstrated very powerful and promising techniques. In this scenario of using deep learning techniques, only one anomaly type can be detected and reported on a single PV module or PV array [6]–[8].

In this article, we propose a technique to solve the problem of pv module multi-defect analyses from drone-captured images of a large-scale solar plant in South Africa. The multi-defect analyses approach helps to successfully detect and recognize various pv module defects from thermographic images and identify the main characteristic anomaly of individual pv modules in an array of pv module strings and substrings. To this extent, we employ the Infrared (IR) Orthogonal images from the UAV-based inspections as our dataset due to the optical and photonics advantage it provides for defects analysis. Our deep learning technique is designed to overcome the challenges experienced by recent state-of-the-art approaches in this research area: Firstly, the need for modification of image size and quality in order to normalize abnormal regions in image features by further cropping and

wrapping of images. In this way, the proposed technique is able to identify and seamlessly extract and differentiate abnormal regions in image features from regular features directly from the original image samples. Secondly, by using a highly advanced CNN extraction model, the proposed technique requires less iterations of processing time for multi-defect PV modules. In this way, the proposed method reduced space utilization and computational resources during training, which further improves time consumption for training and testing. The main contributions of this article are outlined in the below:

- 1) A deep-learning based multiple defects detection technique for solar PV modules on the basis of Infrared Orthogonal images of a 75 MW-ac Solar Plant in South Africa.

- 2) To develop a 112-layered deep neural network with an image input size of 224-by-224 that can classify images into specified number of classes. The network's learned rich feature representation for a wide range of images is leveraged from the pretraining process on different object classes. The Res-CNN3 model is built on a sequence of three identical and concatenated convolutional neural networks and residual blocks which contain skip connections. The skip connections accept the activation from one layer of the neural network and pass it through to another layer deeper into the network, thus, enabling a streamlined and coherent deep-network training process.

- 3) To reduce the number of iterations for processing time of the multi-defect PV modules by using an advanced and modified Faster Resnet-101. The detection process of the detector is improved and equipped with a 3-fold detection framework: Differentiate the most characteristic regions on the multi-defect PV module surface through detection process; mathematically approximate the percentage distribution of all the most distinctive features from the regions differentiated on the PV modules surface; Filter the overlapping abnormal regions, approximate typical features distribution on each of the layers, select the layer with the highest typical features distribution and associate it with one of the differentiated PV module regions.

- 4) Complete the detection process by identifying the most representative region of the multi-defect PV module using a customized classification algorithm.

The rest of this article is consolidated in the following way: Section II discusses the literature review background on PV module anomaly detection with deep learning techniques. Section III outlines the information details about the Solar PV Plant and PV modules specification under investigation in this article, a review of the drone-based inspection system utilized to capture the IR and orthogonal images of the PV arrays, the typical defects of PV modules and the approach proposed in this article for PV module defects detection. Section IV explains the proposed experimental approach. Section V is the experimental evaluation of different techniques compared to the proposed technique in this paper. Section VI concludes the article.

II. BACKGROUND ON PV ANOMALY DETECTION WITH DEEP LEARNING TECHNIQUES

In [9], the authors present an algorithm for rapidly detecting photovoltaic modules containing multiple visible defects by an image analyzing apparatus mounted onto a drone. The proposed algorithmic solution comprises three phases to effectively and precisely analyze different types of module defects. To begin, the Kirsch operator is used to detect anomalous regions, which reduces computational complexity and error rate significantly. Following that, a deep convolutional neural network is used to extract defect features. In the third phase, a multiple classification support vector machine is created to aid in the decision-making process for defect detection. The proposed solution is thoroughly evaluated using a large dataset gathered from real-world solar photovoltaic plants. The experimental results clearly show that the solution is effective for diagnosing photovoltaic modules with multiple visible defects [9].

Karimi *et al.* [10] investigated the defective state of the PV module and created a special Deep Convolutional Neural Network (DCNN) with two convolution layers to classify the relative images. Deutsch *et al.* [11] used a type of DCNN called visual geometry group (VGG) to detect defects in each PV string. Du *et al.* [12] used LeNet, VGG, and GoogLeNet to classify the six types of PV defects (e.g., overheated region). DCNN is a model that is frequently used in these articles. However, because their input image and DCNN network structure differ from object detection, they cannot be used directly for overheated region object detection.

In 'Automated Overheated Region Object Detection of Photovoltaic Module with Thermography Image', the authors propose a Deep Learning-based framework for automatically detecting multiple overheated region targets in PM thermography images. In particular, the presented framework defines scale transform, overheated region detection, and overheated region object detection to be studied. To begin, Skip Connected-DCNN (SC-DCNN) is used to select the thermography images with overheated regions from all of the thermography images collected. Then, Selective Search (SS), Region of Interest-Deep Convolutional Neural Network (RoI-DCNN), and joint learning are used to detect the overheated region object accurately and quickly. Finally, multi-scale training and large-scale testing are used in overheated region object detection to improve performance even further. Categories of experiments are run over the collected dataset to benchmark the proposed framework. It demonstrates that the proposed method outperforms benchmarked methods in terms of effectiveness and efficiency [8].

The study in [13] proposes a method for automatic detection of cracks in low-resolution Electro Luminescence (EL) image data using a deep neural network. The proposed neural network model was trained using large amounts of on-field EL image data collected during the 'All India Survey 2018' from PV power plants across India, including 5 different climatic zones and modules from 20 different manufacturers.

The images are captured by a low-cost CMOS sensor-based camera with a lower resolution that is suitable for field use. The model has been trained and validated for PV cells from various environmental conditions and manufacturers, making crack detection more robust even in lower resolution EL images captured in the field.

Other related techniques which apply the same approach that is proposed in this paper include the destination-driven pedestrian trajectory prediction (DDPTP) technique developed in [24]. This is a deep network architecture that can recognize probable destinations and learn destination-specific trajectories that outperforms existing state-of-the-art models. The Inner-imaging (InI) mechanism proposed in [25] is a technique for modeling the channel relationships in CNNs. It builds inner-imaged maps with both residual and identity mappings, enabling identification flows to participate in the attentional process of residual flows. The InI architecture employs the convolutional grouping filter (G-filter) to organize the grouping relations of the channels, explicitly models the channel in-group coordination and the complementary intergroup ties. This design effectively improves the modeling efficiency of convolutional networks.

III. OVERVIEW OF THE SOLAR PV PLANT SPECIFICATIONS AND DRONE-BASED INSPECTION SYSTEM FOR TYPICAL PV DEFECTS

A. SOLAR PV PLANT AND PV MODULES SPECIFICATION

This is a 75 MW-ac Solar PV Energy Facility (SEF) located in South Africa. The plant consists of 611,100 PV modules in total. The total number of PV modules comprises different quantities of PV modules with different electrical characteristics: there are 2436 PV modules of 125-Watt peak (Wp); 80,640 PV modules with a power max of 130-Watt peak (Wp); 443,100 PV modules specified at 135Wp and 84,924 PV modules with a specification of 140Wp. The total number of PV strings is 43,650. Each PV string consists of 14 PV modules connected in series. The configuration of the PV array table on the other hand consists of 3 PV strings connected in parallel. There are 96 inverters connected in the complete power plant electrical network. These inverters are connected in a total of 48 DC-to-AC conversion cabins, thus, each cabin consists of 2 inverters.

B. REVIEW OF THE DRONE-BASED INSPECTION SYSTEM, TESTBED AND TESTING ENVIRONMENT

The Remotely Piloted Aircraft System (RPAS) Specialized Applications Equipment utilized during the drone flight across the solar plant consist of the following:

- DJI Matrice 200V2 RPAS Drone (Primary of Thermal)
 - Sufficient TB55 Intelligent Batteries
 - Additional WB37 Controller Intelligent Batteries
 - High-Speed SD Cards DJI
- Matrice 210V2 RPAS Drone (Secondary for Thermal)
- DJI Phantom 4 Pro V2.0 RPAS Drone (Primary for RGB)

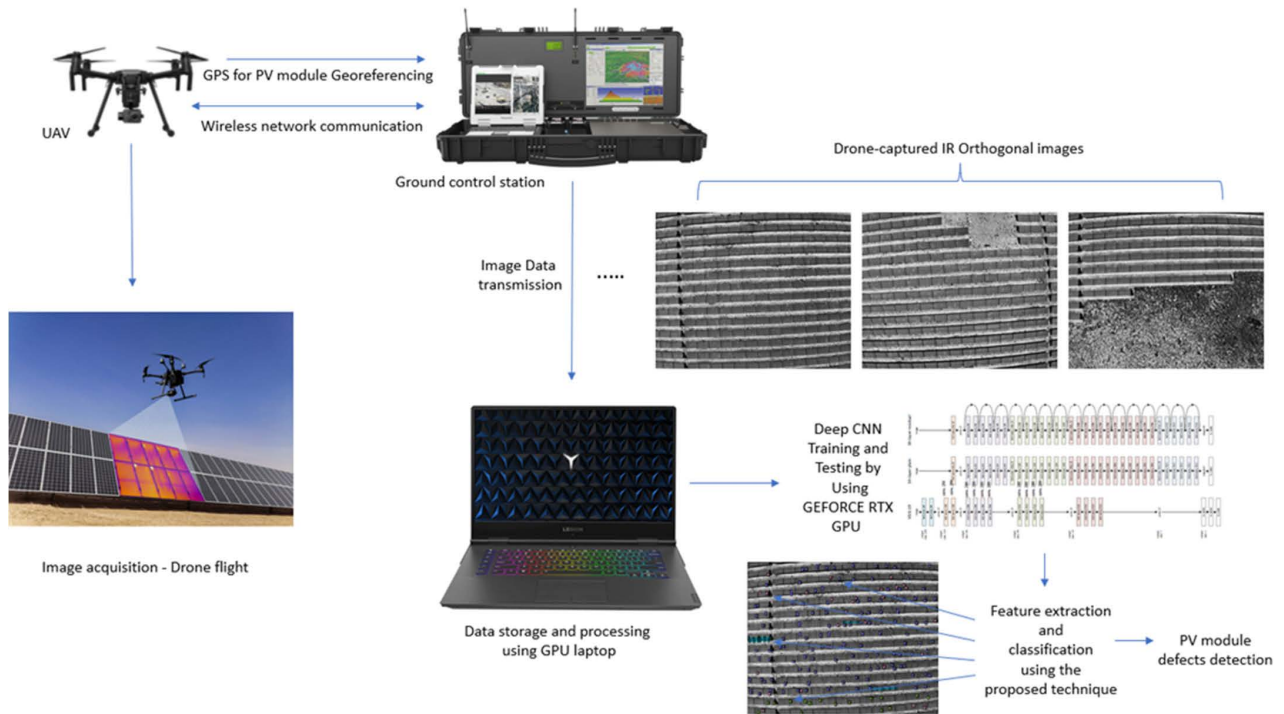


FIGURE 1. Overview of the complete proposed system for UAV-based inspection at the solar plant.

• DJI Zenmuse XT2 Radiometric Thermal Imaging Sensor 19mm 640.

During the inspection, the DJI Matrice 200V2/210V2 RPAS drone is used to fly over the PV plant to collect both infrared and infrared orthogonal images of PV modules and the DJI Phantom 4 Pro V2.0 RPAS drone is used to fly and collect the RGB and RGB Orthogonal images of PV modules. The correlative geographical coordinates of the drones provided by GPS are used to accurately georeference the defective PV modules, which can be retrieved from the DJI Zenmuse XT2 Radiometric Thermal Imaging Sensor 19mm 640 mounted on the drone platform. A 4G wireless communications network transmitter is used to send the imagery data to ground control station (GCS) seamlessly and stored properly. The Ground Sampling Distance (GSD) of the infrared thermal imagery and RGB images collected is based on a level of inspections, which are optimized to detect defects relevant for standard maintenance activities. It provides a balance between speed and accuracy. Relative temperature differentials between cells and modules are used to identify and categorize defects. Level II inspections are able to identify shattering, delamination, module, PID, hotspots and diodes. The RGB imagery makes it possible to improve classifications for soiling and cracked/damaged modules. Figure 2 provides an overview of the complete system proposed for UAV-based inspection for the large-scale PV plant.

The Infrared and RGB images collected by drone inspections were analyzed and sent by the Client. The image analyses were done by utilizing Computer Vision and Artificial Intelligence (AI) technology to identify and localize defects

on PV sites. The analytics engine processes a series of high-resolution thermal imagery without the need to create thermal index orthomosaics and moreover produces the report on the PV anomaly detections based on the software application platform. A level of inspections was performed based on the Ground Sampling Distance (GSD) of the infrared thermal imagery and/or RGB collected.

Furthermore, the infrared and RGB images collected by drone inspections are transferred from the DJI manifold to a workstation used in our testbed. The workstation comprises of a laptop computer with an Intel CORE i7 8th Generation CPU (i7-8750H CPU) @ 2.20GHz, 2208 Mhz, 6 Core(s), 12 Logical Processor(s), a GEFORCE RTX GPU, 16-GB DDR4 RAM and 256 GB NVMe SSD + 1TB HDD. The installed software packages for the execution of machine learning and deep learning experiments are Python 2.7.13 and TensorFlow 1.2.0 and MATLAB.

In this article, only the IR orthogonal images are processed. These raw images are processed by the correlative image processing and analysis functions to characterize the features of PV defects. As there are no publicly available trained models for this activity, the processing of the images is performed in a manual manner by labelling the individual IR orthogonal images containing the PV module arrays to prepare for the image training.

C. TYPICAL DEFECTS ON PV MODULES

Solar PV plant operators can minimize the underperformance of the solar facilities by being proactive in identifying and replacing defective PV modules in the PV fields by following

the best practice preventative maintenance measures. Each of the different types of PV modules manufactured experiences similar and non-similar types of defects. In this article, the typically visible defects of PV modules are discussed. The following section briefly discusses some of these PV module defects:

1) Top glass breakage (along the center): This defect occurs when the top glass of the PV module gets broken along the center of the PV module and across its width. This tends to happen due to incorrect handling and/or misalignment during installation. The defect is generally critical as it may result in the infiltration of water or loss of electrical insulation during normal working condition.

2) Complete module failure: This generally happens when the junction box of the PV module has failed. The failure of the junction box results in the complete PV module being electrically non-functional. An electrically non-functional PV module measures 0 volts when tested for open-circuit voltage under irradiant conditions that meet the standard or minimum irradiance for PV module electrical generation.

3) Burn marks: Burn marks occur on the top glass surface of the PV module. These burn marks have a variation of colours from white, yellow, and orange to red. Burn marks could be caused by defective bypass diode and/or by the formation of hot spots. Further burning of the module could result in leakage current and loss of electrical insulation during normal working conditions.

4) Encapsulant delamination: This occurs as visible delamination between encapsulant (himalan) and back glass near the terminal box. The adherence strength loss among the PV module's different layers between the encapsulant (EVA) and the solar cell could result in delamination. The appearance of himilan delamination does not cause a reduction in electrical output performance or reliability of the product [15].

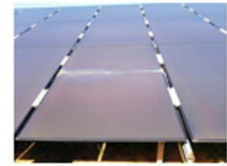
5) Cracking (or back glass breakage): This appears as glass broken/scattering starting from any part of the edge. Cracking is caused by chipping both on edge and/or any other part of the back glass, overtightening on the module clamps, worn-out rubber or lack thereof within the module clamps and abnormal or misaligned PV steel structure configuration. The variation of the environmental conditions, such as wind and snow load, brings a higher static mechanical load, which causes glass breakage or results in the scattering phenomenon of the cracks, due to thermal expansion and contraction in normal working operation.

D. APPROACH FOR DEFECTS DETECTION

1) PROPOSED MODEL FOR FEATURE EXTRACTION

The proposed network architecture for PV module features extraction is the Res-CNN3. Res-CNN3 is a CNN which is 112 layers deep and can classify images into a specified number of classes. The pretrained network can classify images into 1000 object categories, such as keyboard, mouse, pencil, and many animals. As a result, the network has learned

Top Glass breakage



Complete PV module Failure (shows no physical defect)



Burn marks



Delamination



Cracking



FIGURE 2. Review of images of PV modules with different defects.

rich feature representations for a wide range of images. The network has an image input size of 224-by-224. Res-CNN3 comprises a sequence of three identical and concatenated convolutional neural networks as well as residual blocks which contain skip connections. The skip connections accept the activation from one layer of the neural network and pass it through to another layer deeper into the network. This makes it possible to train deeper networks with simplicity. This CNN architecture forms the foundation of the proposed feature extraction encoder module.

Convolutional layers typically include 3×3 filters and adhere to two basic design principles: (1) for the same output feature map size, the layers have the same number of filters; and (2) if the feature map size is half, the number of filters is doubled to maintain the temporal complexity per layer. Direct downsampling is also performed by means of convolutional layers with a stride of 2. The network concludes with a global average pooling layer and a 1000-way fully-connected layer with softmax.

2) DEFECT CLASSIFICATION DEEP NEURAL NETWORK

The main contributory element in the defect classification network is founded on the VGG-19 component of the Resnet101 [16]. From the second convolutional layer (conv3-64) we multiply the number of channels by a factor of three for the third convolutional layer (conv3-256) and iterate the same width or number of channels through three times over two sequential max-pooling layers. This design helps us to process the PV module image multiple times through the same number of channels, thus, improving efficiency in handling complex features and accuracy in differentiating the thermal delta across the PV module image, whilst maintaining minimum computational resource demands on the training and testing platform.

3) DEFECTS REGIONS DETECTION OF PV MODULE

This process of defects regions detection of PV module image has a bipartite approach that identifies whether the PV module image contains thermal delta representative of PV module defects or not. This serves as a sample for more detailed and categorical defects region object detection. As a result, defects regions detection of PV module has to take place first. The careful design and modification of the proposed model for feature extraction and defects classification refine the anomaly detection process on the PV module images. The first layers of the proposed feature extraction model are intended to extract the features of the image with the defective PV module.

As the drone flight across the PV plant is such that the drone-captured images are representative of arrays of PV modules instead of individual PV modules, individual PV module objects do not typically occupy the whole image, rather multiple PV module objects connected across PV arrays in PV strings and PV substrings do. This characteristic of the dataset inspires the design of the deep convolutional neural network for defects regions detection of PV modules. Instead of a reliance on computation over multiple scales due to some multi-scale features, and stacking due to objects occupying the whole image, our reliance is more on averaging and max-pooling.

The first 12 weight layers only comprise of 2 different scales, starting with a scale of 64 in the first 2 layers and multiplying weight layers 3 up to 12 by a factor of 4 and thus creating the same width across 10 weight layers. Layers 13 up to 15 are then multiplied by a factor of 2. The multi-scale training setting or training set augmentation process thus allows for only 3 different scales, this is a rather small range of scales for object recognition and speeds up the training process and reduces the fine-tuning of layers. The rest of the model contains parameter layers for the residual network. The final stage of the residual network classifies the extracted features and comprises of an average-pooling layer and a single fully connected (FC) layer. The loss function that is employed is a logistic regression log ($\cdot$). The objective

function can thus be explained in the following manner:

$$L^d(\{x_i^d, y_i^d\}; W^d) = y_i^d \log(p(y_i^d | x_i^d; W^d)) \quad (1)$$

$p(\cdot)$ means the probability of the corresponding model.

4) DEFECT REGION OBJECT DETECTION OF PV MODULE

After the defects regions detection of PV module has taken place. The more detailed and categorical defects region object detection follows. To achieve this, Selective Search (SS) algorithm [17], [18], is employed to place several RoIs (Regions of Interest) in the thermal image of the PV module. The defects region object detection model will then detect the thermal region object together with the thermal image and the RoIs. In the end, the classification and location of each RoI are determined and predicted in the final output of the proposed technique. Generally, the scale of the complete thermal image of the PV module is much larger than the scale of the defective regions of the PV in the thermal image. The Selective Search (SS) algorithm predicts the RoIs in the input PV module thermal image. The defects of the PV module by means of a thermal delta are located within the area of these RoIs. In this way, Selective Search is very advantageous for defects region object detection. Each RoI is depicted by a Bounding Box by means of the region proposal coordinates of the SS. The SS equation is defined below:

$$\{S_j^k\}_{k=1:K} = (S_j^1; S_j^2; S_j^3 \dots S_j^K) = f^{SS}(x_j^{od}) \quad (2)$$

It is only the most precise RoIs, which are selected as being representative of defective regions of the PV module thermal image. This selection is determined by the intersection over union (IoU) with their labelled defective PV module region target. If the area of the defective region object overlaps approximately 30% to 40% with the proposals RoIs in IoU ($40\% \geq \text{IoU} \geq 30\%$) then the RoIs can be selected to be input with the thermal image. Conversely, when the IoU is more than 40%, the proposal RoI is denoted as the positive sample for the defective region object detection.

IV. PROPOSED EXPERIMENTAL APPROACH

A. DATASET DETAILS AND IMAGE PROCESSING

The thermal images of the PV modules were collected from a 75 MWp solar PV plant located in Free State Province, South Africa. The normal thermal operation range of the PV modules is approximately 28°–55° C. All the thermographic images are captured by the DJI Zenmuse XT2 Radiometric Thermal Imaging Sensor 19mm 640 mounted on the DJI Matrice 200V2 RPAS drone and the DJI Matrice 210V2 RPAS drone between 9:30 AM and 3:00 PM in the clear days of November 2021.

All the images are captured in 640×512 scale and contain formats of RGB, infrared and orthogonal images. The selection of the IR orthogonal images is based on the analyses of the behavior of infrared light and irradiation through glass surfaces [28]. This behavior aids in the analyses and observation of anomalies on the front surface among glass matrix,

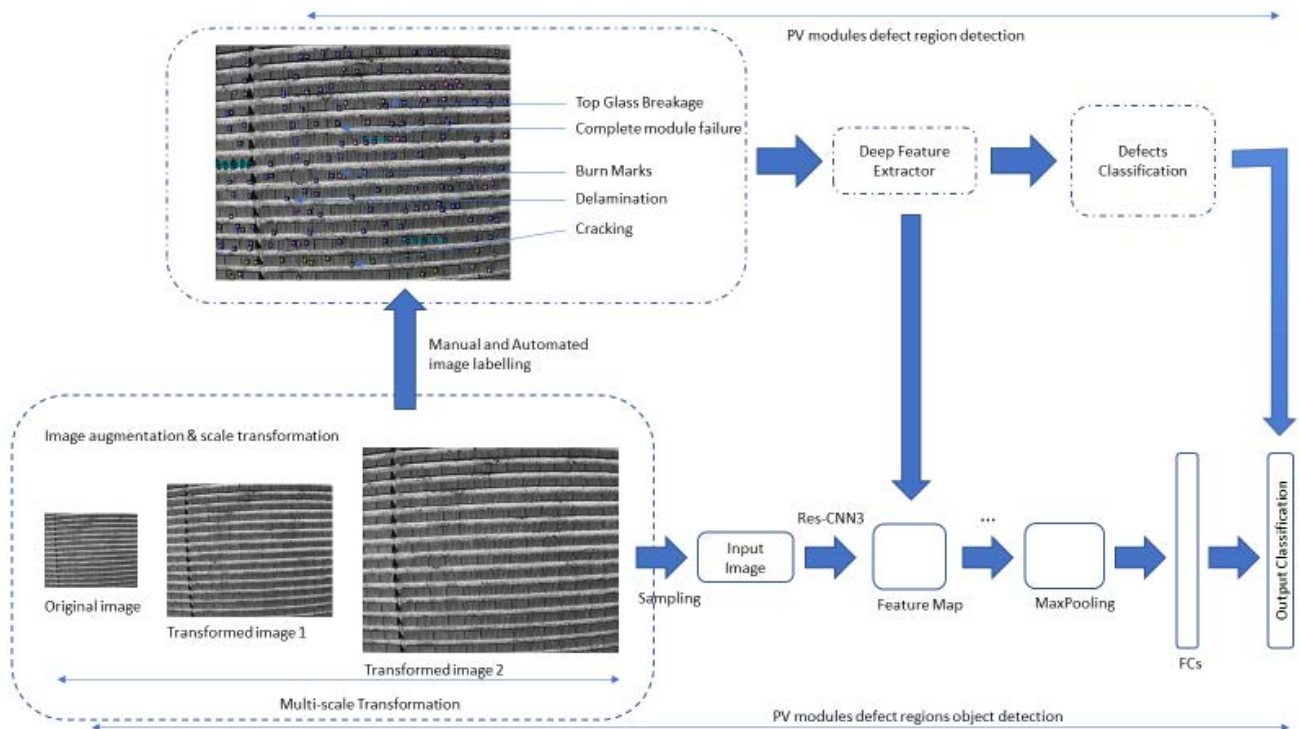


FIGURE 3. Framework of Res-CNN3 for defective PV modules region object detection.

which is a good application in optics, laser and photonics. The total number of these thermographic images is 14892. All the images contain multiple PV arrays or PV subfields showing PV modules with different PV module defects. The level of inspection adopted for the drone inspections is the most detailed and conforms to the IEC TS 62446-3 thermography standard. The high GSD is able to accurately capture the absolute temperature values of solar cells. This can be used to support warranty claims for defective PV modules, diagnose systemic cell level defects, and identify small cracks in modules.

B. EXPERIMENTAL SETUP

During the experiments, there are a few deep learning techniques, which are tested and assessed for performance comparisons as part of this article. Amongst the selected techniques is the Service Provider method, which is not identified by name due to the company's proprietary information rights, this is a deep learning model, and is referred to in this article as 'SP method'. There are other methods used in the experimental setup and these are listed next. The Inception-v3 [19], is an advanced and well-applied method that trumps most of the inception networks. DarkNet-53 is a deep learning algorithm that is beginning to gain traction in applications for highly advanced object detection problems and it has very effective detection accuracy [20]. InI-WRN-16-8-square-3 is an application of the InI module on the wide residual network backbone, which comprises square G-filters for CNN structure modelling [25], [26]. InI-PyramidNet-mix-5 is a result-

tant CNN model from the application of the InI module on the pyramidal residual network backbone [25], [27], it comprises mixed G-filters for structural modelling of CNNs. Additionally, GoogleNet [21] and YOLO [22] are experimented with and assessed for PV module anomaly detection.

The selected deep learning models are well-trained according to individual model performance. Every technique is trained and tested five times and the average performance is reported to ensure certainty about the accuracy of the results. The main anomaly types or PV module defects to be detected during experiments are Cracking, Encapsulant delamination, Complete module failure, Top-glass breakage and Burn marks. The PV module defects manual labelling is done in MATLAB, automated labelling process is then performed based on the results of the manual labelling process. All the experiments are performed in python 2.7.13 environment of the Lenovo Legion Y740 Laptop on Windows 10 with Intel CORE i7 8th Generation CPU (i7-8750H CPU) @ 2.20GHz, 2208 Mhz, 6 Core(s), 12 Logical Processor(s), a GEFORCE RTX GPU, 16-GB DDR4 RAM and 256 GB NVMe SSD + 1TB HDD.

C. EVALUATION METRICS

The evaluation metrics selected for assessing the performance of the experimented methods including the proposed method are precision (P) and recall (R). For the defective PV module region object detection, the efficiency must be evaluated and analyzed carefully. In this article, Average Precision (AP) serve as the 11 interpolated points on precision and recall

curves [17]. Furthermore, it is employed for the performance assessment of associated tests with Intersection over Union (IoU) $\geq 50\%$ for the proposal Bounding Box (BBox) and labelled BBox. Additionally, the average test time of each image (t^{ATE}) is used for evaluating the effectiveness of defective region object detection. The formulae of the evaluation metrics utilized in the experiments are defined below:

$$t^{ATE} = t^e / K^{te} \quad (3)$$

where t^{ATE} is described by the total test time (t^e) and iterations K^{te} .

$$P = \frac{TP}{TP + FP} \quad (4)$$

where TP is True Positives and FP is False Positives.

$$R = \frac{TP}{TP + FN} \quad (5)$$

where FN is False Negatives.

$$AP^{IoU \geq 50\%} = 1/11 \sum P_{smooth}(k) | IoU \geq 50\% \quad (6)$$

D. PARAMETERS SETTING

The parameters setting of the proposed model follows a very specific sequence, which is determined as part of the article. Stochastic gradient descent is used to fine-tune the proposed model with a batch size of 16 sample visible images of known PV module defects. The learning rate, momentum, and weight decay are 0.03, 0.88, and 0.001, respectively. The momentum is utilized to speed up the convergence of the designed loss function [23]. For reduction of the model's training error, the model needs and utilizes the weight decay given a small assigned value. After 700 epochs, the training process of the proposed model stops as the accuracy is no longer increased.

V. RESULTS AND DISCUSSION

As part of the performance assessment of each model, we initially, analyze the number of parameters of each model compared to the training time on the same machine with the same training dataset. This is an important evaluation as it helps us to establish the efficiency of each model in terms of training time and required computing resources. The next line of performance evaluation for each deep learning model focuses on the average precision of individual deep learning models as regards different PV module defects region object detection. This is a very elaborate and precise assessment process, for each deep learning model we analyze and assess the accuracy of detecting individual PV module defects on testing data after being trained on a sample training dataset.

During the experimental process, we also transform the scale of the PV modules into different scales in order to identify the best scale at which most of the deep learning

models are able to efficiently extract features of the images and generalize well about the PV module defects. The criterion for this assessment incorporates, average precision $AP^{IoU \geq 50\%}$, average testing time t^{ATE} , single-scale transform and multi-scale transform. Finally, we evaluate the individual deep learning techniques' performance on defective PV module region detection. This area of testing and performance evaluation has to do with the ability and accuracy of the deep learning models to detect, which of the PV modules in the PV arrays presented in the thermal images have some form or type of defect, even if the model might not be accurate about the type of defect present in the PV module or PV array.

We examine the influence of the number of parameters of each deep learning model on the training time of each model. In general, the models that take the least time to train whilst containing many parameters useful for detection accuracy can be described as highly efficient. This is because a lower cost GPU or high-performance computing platform can be effectively utilized to execute the model for different applications.

The objective of this section is to analyze and assess model efficiency based on number of parameters and training time. These analyses and assessments do not include a detailed analyses of how learned features are processed and transferred by each deep learning model's CNN layers due to individual model design and architecture resulting in the training duration experienced. This detailed CNN layers' feature-processing analysis is only done for the proposed model.

Figure 4 illustrates the number of parameters of each of the experimental deep learning models including the proposed Res-CNN3 model. YOLO has the highest number of parameters followed by the Res-CNN3 model. GoogleNet contains the least quantity of parameters followed by the InI-WRN-16-8-square-3 model, which contains 11,14 million parameters. Although Res-CNN3 contains the second-highest number of parameters across all the different tested models, it only takes the third shortest time to train, this is characteristic of a very efficient and practical model.

GoogleNet takes the shortest time to train, however, it contains the lowest number of parameters across the board. Res-CNN3 on the other hand contains a very high number of parameters, the second highest, yet it takes a relatively and comparatively short time to train. The main enabler of the rapid training time for the Res-CNN3 model is the iterated multiplication factor of weight layers across multiple weight layers. This iteration results in the same width across multiple weight layers. A reduced range of scales conversely improves the training speed of the Res-CNN3.

For the following part, we compare our proposed model with the other advanced deep learning methods: YOLO, GoogleNet, Inception-v3, DarkNet-53, InI-PyramidNet-mix-5, InI-WRN-16-8-square-3 and SP Method on the accuracy of detecting different PV module defects.

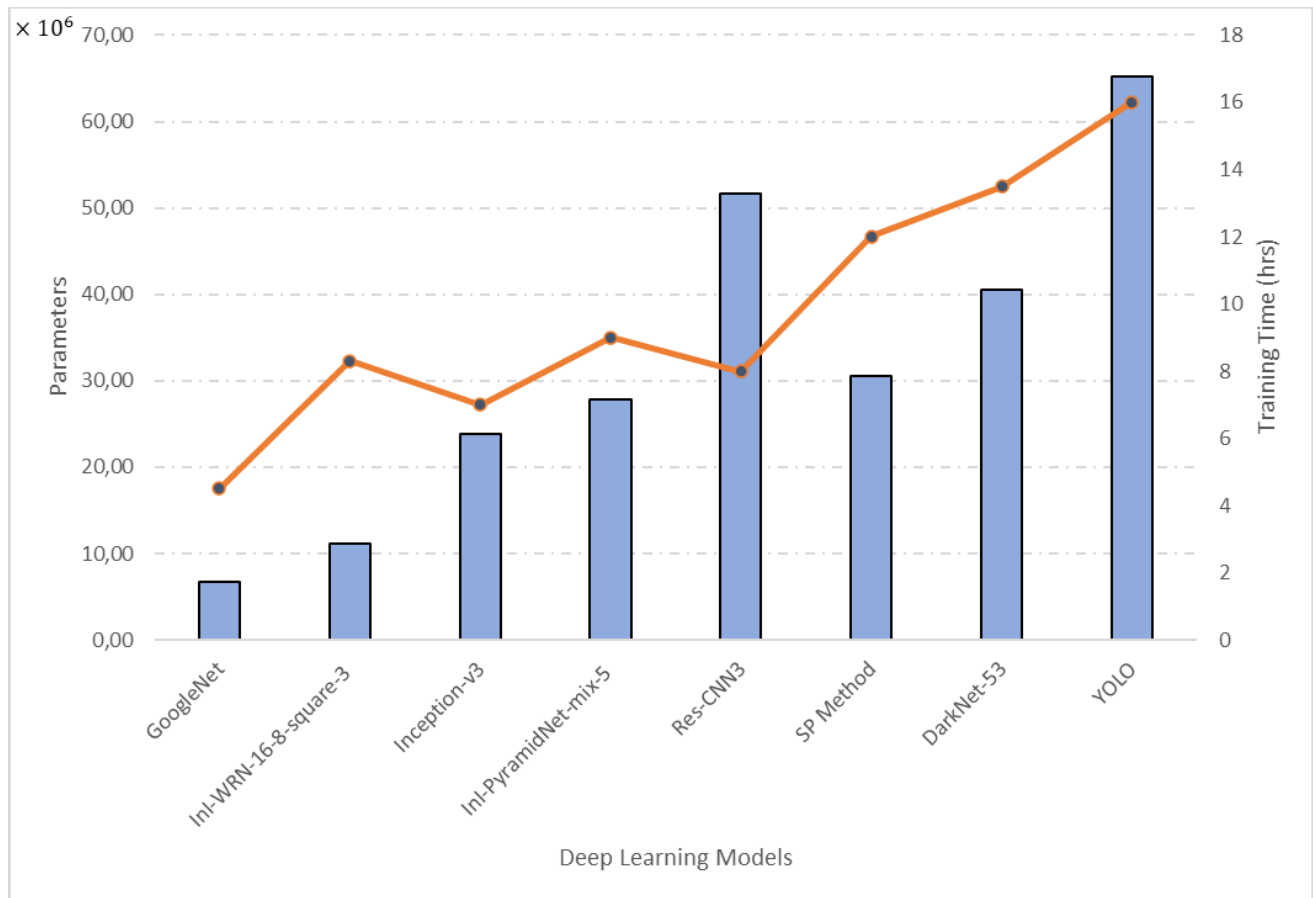


FIGURE 4. Comparison of parameters and the training time of each model.

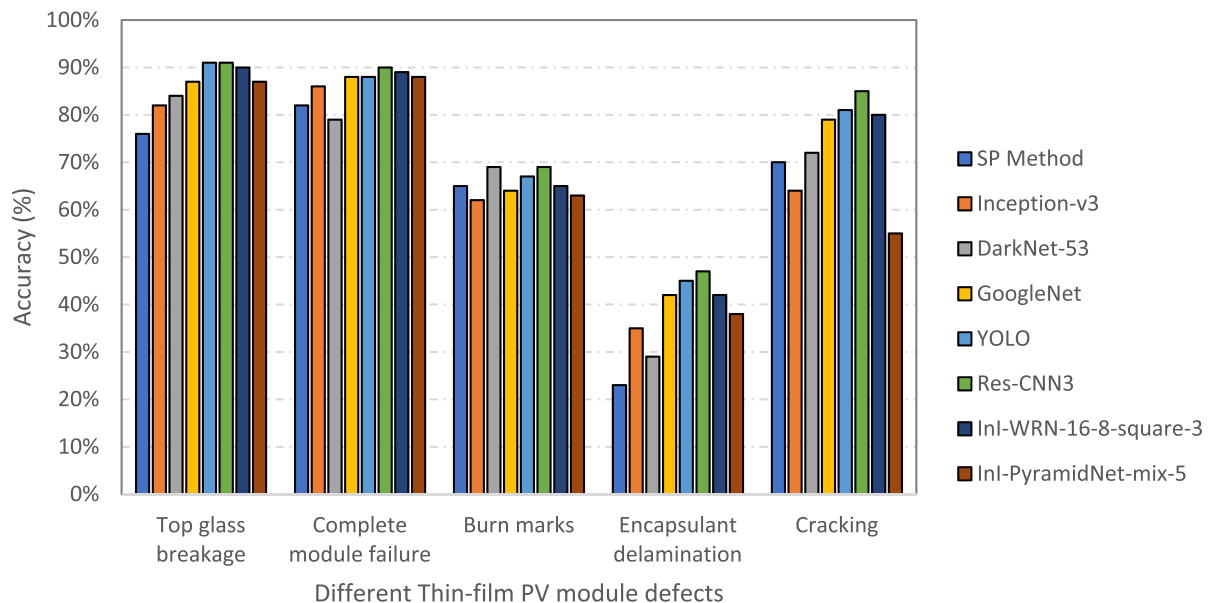


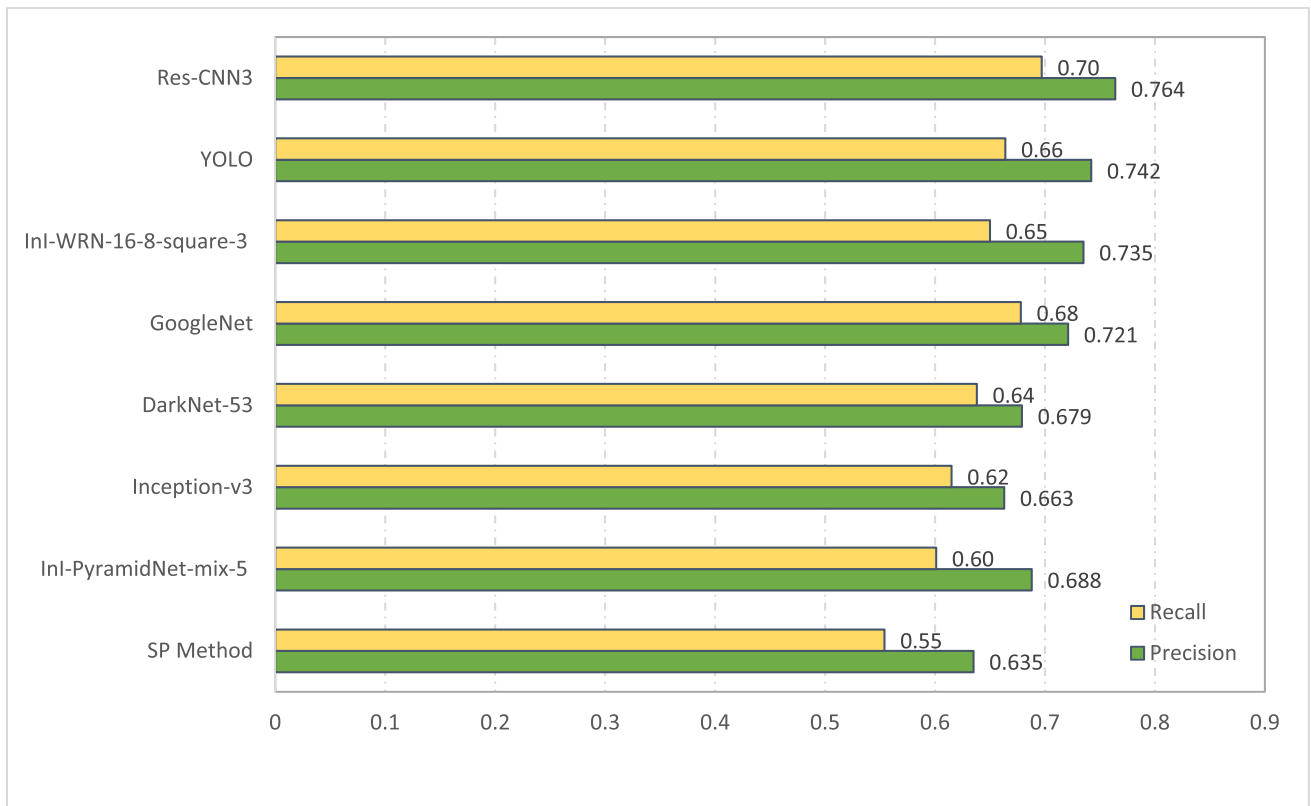
FIGURE 5. Average Precision (AP) of the proposed model compared to other state-of-the-art models for detection of different PV module defects.

Mean average precision is used as the evaluation metric for this performance assessment and the results are shown in Figure 5 above.

As can be seen in Figure 5, most of the models perform better in the detection of Top glass breakage, Complete module failure and Cracking. In all these scenarios, Res-CNN3

TABLE 1. PV module defects region object detection with different image scale transforms.

Training and Testing scales	SP Method	Inception-v3	Res-CNN3	DarkNet-53	GoogleNet	YOLO	InI-WRN-16-8-square-3	InI-PyramidNet-mix-5
640×640	64.4%/0.054s	53.3%/0.025s	64.5%/0.049s	56.3%/0.063s	51.6%/0.037s	58.3%/0.049s	62.4%/0.051s	64.9%/0.051s
520×520	73.5%/0.023s	58.5%/0.039s	67.8%/0.052s	78.1%/0.027s	63.2%/0.066s	69.7%/0.019s	64.3%/0.059s	75.9%/0.022s
480×480	59.3%/0.037s	61.4%/0.043s	59.4%/0.051s	54.1%/0.048s	57.5%/0.073s	63.2%/0.052s	56.1%/0.069s	61.7%/0.041s
640×520	66.4%/0.047s	74.3%/0.063s	75.2%/0.052s	71.6%/0.057s	63.2%/0.066s	67.5%/0.073s	73.5%/0.063s	68.4%/0.042s
640×480	67.9%/0.035s	78.1%/0.027s	81.7%/0.029s	75.7%/0.032s	73.7%/0.036s	69.9%/0.038s	74.2%/0.063s	67.9%/0.034s
520×480	56.7%/0.031s	63.7%/0.039s	76.7%/0.069s	67.9%/0.035s	61.7%/0.057s	54.1%/0.048s	73.1%/0.071s	56.7%/0.036s
640×520×480	63.5%/0.063s	54.6%/0.051s	66.9%/0.061s	62.3%/0.056s	64.7%/0.076s	49.7%/0.069s	59.3%/0.071s	52.9%/0.078s

**FIGURE 6.** Deep Learning techniques comparisons on defective PV module detection.

achieves the best performance. For Cracking and complete module failure, Res-CNN3 has the highest accuracy of detecting these individual PV module defects. In the scenario of ‘Top glass breakage’, YOLO and Res-CNN3 achieves the same highest accuracy for this defect detection.

The mean average precision of Res-CNN3 across all PV module defects is 76.4%. The main reason for the lower mAP is that all the models battle to detect burn marks and encapsulant delamination. Nevertheless, for both Burn marks and encapsulant delamination, Res-CNN3 achieves the highest accuracy. We find that the performance of all the tested models significantly improves on the detection of burn marks and encapsulant delamination defects when the thermal image

scales are transformed and augmented. As such, the challenge of detecting these two defects is not necessarily inherent in the deep learning models themselves, however, it is in the feature extraction of different PV module defects on thermal images with different scales.

In the next section, the scales of the thermal images used for training and testing are adjusted or transformed in a singular fashion and as well as multi-dimensional fashion. The objective of this single scale and multiscale image transformation is to evaluate what influence the scale of the images has on the detection accuracy $AP^{IoU \geq 50\%}$ and total test time t^{ATE} . The original scale of the images after drone inspections is 640×512 . The original images are augmented and

transformed into multiple scales and then used for training and testing. Various single-scale and multiscale transformations are applied to the PV module images for defective PV module region object detection. The results of each model after training and testing on the different image scales are presented in Table 1.

The general AP results for each model become better as the single scale of the training and testing images increases. The performance of $AP^{IoU \geq 50\%}$ converges at approximately 640×520 and 640×480 image scales. What is also notable is that only AP increases as we get to a higher image scale on the single plane of scaling. The average testing time for each model is rather varied irregularly across different models based on the image scaling. In particular, Res-CNN3 produce the best $AP^{IoU \geq 50\%}$ results on the 640×480 image scale followed by Inception-v3. However, in terms of t^{ATE} , Inception-v3 takes the lead in achieving the fastest average testing time at 0.027s on the same image scales followed by Res-CNN3 with a t^{ATE} of 0.029s. The next best performing model is DarkNet-53 with a performance of 75.7%/0.032s on the 640×480 . All the results in this area of performance assessment are based on the models being exposed to the same image scales during training and testing.

For multiscale training and testing, the models are trained and tested with only a single type of scaling - $640 \times 520 \times 480$. What is immediately observed is that the training and testing on this type of scaling takes a longer t^{ATE} for each model compared to the single scale. This is expected because multiscale training and testing takes more memory. When considering the weight of each evaluation metric in the results, for both types of scaling, it is clear that Res-CNN3 is more robust for training and testing and more pragmatic in implementation.

The main criteria of performance evaluated here is the accuracy of defective PV module region detection without a particular focus on any type of defect present in the thermal image. There are only two different classes relevant in this instance: normal PV module and defective PV module. The scale of all the training and testing images is 640×512 . All the deep learning models perform very well in this aspect. These test results are benchmarked on Recall and Precision to assess the individual deep learning models. Figure 6 presents the results of the tested deep learning techniques on the detection of defective PV module regions.

Res-CNN3 achieves the best results in this scenario. What is interesting to observe is that the results of defective PV module region detection are not in any way dependent on the model depth or number of parameters. All the tested deep learning models YOLO, InI-WRN-16-8-square-3, GoogleNet, DarkNet-53, Inception-v3, InI-PyramidNet-mix-5 and SP method perform differently in this criterion of performance assessment. This outcome of results is also different to the mean average precision (mAP) of each model for detecting different PV module defects based on the type of defect present in the PV module thermal image.

The underlying recognition, however, is the potent efficiency, robustness and practicality of Res-CNN3 in defective PV module detection and PV module defective region object detection.

VI. CONCLUSION

PV module field anomaly detection from drone-based IR and RGB orthogonal images is a very efficient and critical operation and maintenance strategy for solar photovoltaic plants. There are few research works on solar PV plant maintenance using drone inspections and solar PV defects detection. The area of research is, however, still developing on how deep learning methods can be successfully applied to detect multiple and various PV defects from drone-captured images of solar PV plants. This article proposes and presents in a detailed fashion a deep learning-based technique, which can detect various PV module defects of PV modules from IR orthogonal images captured during a drone flight. The technique applies a bipartite process for defects regions detection to process the thermal delta of a thermal PV module image using a combination of concatenated CNNs and residual networks to identify whether the thermal image is characteristic of defect regions or not. Logistic regression is employed as the loss function. Selective Search (SS) is also applied for the PV module defect regions object detection to differentiate anomalies. The proposed technique can seamlessly differentiate between PV modules that are defective from those that are not characteristic of any defects. The technique is also evidently both fast during training and testing and is very pragmatic to implement in different applications as it is a low-cost resource-consumption model, computationally.

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system of deep learning methods in computer vision and renewable energy systems.



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